STAT GU4204 Spring 2025 HWs

Please make sure you read the instructions below and the syllabus section on use of AI in CourseWorks carefully.

HW will be submitted online in CourseWorks as a **single** **.pdf** file **before** the due date/time.
Late HW is not accepted. If CourseWorks marks the HW late, the HW will be treated as late.

If HW is submitted within the grace period of two hours following the deadline, -1 point penalty applies. HW submitted after the grace period of two hours after the deadline is not accepted.

Students are advised not to leave HW submission to the last minute since if something unexpected happens during HW submission while submitting the HW last minute, it results in lost points.

Requests for late HW to be accepted, late penalty points not to be deducted, or for handing in the HW late cannot be honored (unless you have a documented disability, verified through disability services, that provides such accommodation).

Lowest HW score will be dropped.

All submitted HW needs to be hand-written (not typed, hand-writing on a tablet is fine) unless you have a documented disability, verified through disability services, that prevents this; or unless otherwise stated in the HW. You can also hand-write it on a tablet.

New HW will be added to this document.

You are encouraged to discuss HW problems with classmates. However, each student should write their solutions in their own words. If you discuss any HW problem with classmates, you need to write clearly on the first page of that HW which questions are discussed with which classmates (names, last names, UNIs).

Show all work to get full credit. Calculators can be used in the HWs but NOT in the Exams.

As HWs will be graded based on submission/effort, once the course TA posts the HW solutions, it's the students' responsibility to compare their answers to HW problems with the posted solutions and learn from their mistakes.

As per department rules, students should not be contacting the grader. For any HW grade objections students should contact the course TA. Grade objections to any HW can be done within the week following the posting of the solutions to that HW. Grade objections after that time cannot be honored.

Undergraduate Statistics Help Room:
The Department of Statistics offers a help-room service for students enrolled in undergraduate Statistics courses. The purpose of the help-room is to provide peer-to-peer tutoring, and to facilitate the formation of study groups. For most current schedule see Help room webpage http://stat.columbia.edu/help-room/ .

Course TA: Baichen Tan, bt2641@columbia.edu
OHs: M 4-7pm, W 10am-12pm.
For course TA's office hour location and most up to date information on TA office hours/location/format see announcements that will be made on CourseWorks.

All course related questions should be first directed to your course TA as we talked about it in class.
If multiple students ask the same question, an announcement through CourseWorks will be sent.
You should make sure that your CourseWorks settings is set to receive these announcements.

HWs are listed on the next page.

©Dr. Banu Baydil, 2026

HW1 (**Out**: 02/27, **Due**: **02/06, 3pm**)

1) a) Solve the following exercises from the course textbook and submit them as part of your HW:

Ch 6.3: #14a;

Ch 7.5: #2, 6;

Ch 7.6 (For exercises in this section: i) If needed, you can use the results you found in 7.5 exercises above or in 1) below without recalculating them, ii) In addition, if needed, for 18-20, you can use the results in the textbook or the lecture notes without recalculating them): #2, 6, 12, 18, 20.

2) Solve the following exercises from the lecture notes and submit them as part of your HW:

a, b) Lecture Slides STAT_GU_4204_03 pg. 9: Exercises 1, 2;

c) Lecture Slides STAT_GU_4204_04 pg. 18: Exercise 2.

Additional assigned exercises (**Out**: 02/27)**:**

1) The solutions to the following <u>assigned</u> exercises from the course textbook will be posted, however, they will not be counted towards grading of the HW, hence you do not need to submit them as part of your HW:

Ch 6.3: #13, 15;

Ch 7.5: #3, 5a, 7, 9;

Ch 7.6 (For exercises in this section: i) If needed, you can use the results you found in 7.5 exercises above or in 1a) above without recalculating them, ii) In addition, if needed, for 19-21, you can use the results in the textbook or the lecture notes without recalculating them): #3, 19, 21.

2) The solutions to the following <u>assigned</u> review exercises for probability background will be posted, however, they will not be counted towards grading of the HW, hence they do not need to be submitted as part of your HW. (These <u>assigned</u> review exercises will continue in the next HW as well). Students are responsible from knowing, and are expected to know, how to solve the below types of problems:

a) The following exercises from the course textbook:

Ch 4.6: #10, 13, 14 (Hint: For exc. #14, you can use the result in exc. Ch4.6 # 8 without proving it. Also recall Cov(W,W)=Var(W));

Ch 6.1: #2 (Hint: Corr. 5.6.2, Th'm. 5.6.6);

Ch 6.2: #1, 3, 5, 7.

b) Proving Th'm 5.7.3 from the textbook using the change of variables given in its proof.

c) Using definition of expected value, computing $E(X^3)$ where X has gamma distribution. (Hint: Proof of Th'm 5.7.5 and your result in part 3b) above).

d) For each case below, computing the moment generating function of $X_1+X_2+X_3+X_4+X_5$ and from this identify identifying the distribution of $X_1+X_2+X_3+X_4+X_5$ when X_i, when X_i, i=1, …,5, are independently sampled from:

a) Bernoulli(p);

b) Binomial(n_i, p);

c) Gamma(α_i, β).

(Unofficial (not collected) HW2) Additional Assigned Exercises (**Out:** 3/27):

The solutions to the following <u>assigned</u> exercises from the course textbook will be posted, however, they will not be collected, hence you do not need to submit them:

i) Solve the following exercises from the course textbook:

Ch 7.7: #1;

Ch 8.2: #9;

Ch 8.4: #3.

ii) b) Lecture Slides STAT_GU_4204_06 the last exercise on pg. 41.

©Dr. Banu Baydil, 2026

HW3 (**Out**: 04/22, **Due**: **04/28, 3pm**)

1) Solve the following exercises from the course textbook and submit them as part of your HW:

Ch 8: 8.5: #2a, 4;
8.7: #11a;
8.8: #2, 7, 17;
Ch 9: 9.1: #1a.

HW4-Extra Credit (Out: 05/02, Possible to submit until 05/05, 3pm, with grace period until 05/05, 5pm):
Note: This HW is on most recent material covered after the last HW. Students **are responsible** for the final exam from the problems in this HW.
<u>However, submission of this HW is optional and is for extra credit</u>. Its grading will be based on submission and effort. Solutions will be posted. If a student does not submit EC HW, it will not negatively affect their course grade. Effect of EC HW will be taken into account after class average is computed and course grades are assigned.

i) Solve the following exercises from the course textbook.

Ch 9: 9.5: #2, 4;
9.6: #1, and the following additional question:

Q1) Consider the pdf and the hypothesis test given in the textbook in exercise 9.10: #4. Given this hypothesis test rejects H_0 when $x>1/3$, compute probabilities of Type I and Type II errors.

©Dr. Banu Baydil, 2026